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THEORETICAL ARTICLE

Optimizing a sustainable logistics problem in a renewable energy network using a genetic algorithm

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Abstract

Renewable energy sources, including bio-energy technologies, have been introduced to overcome sustainability challenges, such as negative environmental impacts and energy insecurity due to reliance on fossil fuels. Logistics activities have a significant effect on the cost and environmental impacts of renewable energy supply chains. Understanding and reducing the carbon footprint of renewable energy supply chains can aid in mitigating environmental impacts. Thus, this research presents a mathematical model that can be used to optimize renewable energy supply chain logistics costs and carbon footprint. The proposed model considers a biomass-to-bio-oil supply chain, including harvesting and collection sites, bio-refineries, and distribution centers. It is assumed that mobile and fixed refineries will be used to produce bio-oil. The model considers the mass of biomass and bio-oil, number of mobile and fixed refineries, and number of truck trips to minimize total cost, where a carbon tax is used to represent carbon footprint in the mathematical cost model. A genetic algorithm is designed to obtain a near-optimal solution. Six scenarios for mobile and fixed refinery capacity are tested in performing sensitivity analysis of the model. The results indicate that the mathematical model of the bio-oil supply chain has reasonable relationships between input and output variables. The model is able to incorporate the impact of carbon emissions in a mixed-refinery bio-oil supply chain as a cost parameter. It was also found that increasing mobile refinery capacity has the greater effect on reducing total cost and carbon emissions than increasing fixed refinery capacity.

Keywords Sustainable supply chains management · Carbon tax · Genetic algorithm · Mixed integer linear programming model

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1 Introduction

In recent years, supply chain management (SCM) principles have been applied to reduce total production system cost. SCM integrates several parts of a logistics-manufacturing network to obtain an optimal or near-optimal decision. It is vital for decision makers to simultaneously consider all parts of a network due to trade-offs that may occur when considering parts of the system independently. Warehousing, for example, desires to hold lower product inventory levels to reduce holding costs, while sales attempts to have higher inventory levels to avoid losses of sales. SCM aids in managing the flow of goods in a system by assisting in decisions about production, inventory, location, logistics, and delivery with respect to constraints on the system.

With a growing call for sustainable industrial development, which is as an approach to provide society with responsible products, SCM principles have been applied to address existing barriers, such as resource constraints and competing industrial demands. A surge of interest can be seen in assessing the sustainability performance of supply chain activities [1]. As a part of sustainable industrial development, sustainable manufacturing has been described by the U.S. Department of Commerce as, "creation of a manufactured product with processes that have minimal negative impact on the environment, conserve energy and natural resources, are safe for employees and communities, and are economically sound" [2]. A particular area of interest over the past half century has been in reducing the use of fossil fuel-based energy, due to concerns over energy security and environmental impacts of extraction, processing, and use, among other reasons [3].

The burning of fossil fuels by humans has a critical role in changing the proportions of atmosphere gases leading to global warming. Renewable energy sources (e.g., bioenergy, solar energy, and wind energy) have been explored as alternatives to fossil-based energy for the past several decades [4]. According to the *REN21 Renewables Global Status Report*, little progress has been made, with renewable energy only comprising 19.3% of global energy consumption in 2017 [5]. While a limited number of countries have readily accessible fossil fuel energy sources, many regions are able to provide renewable energy resources. Thus, the need for efficient, low-cost, and environmentally-responsible production of renewable energy has attracted considerable attention globally.

Bio-oil is considered a source of clean, renewable energy [6]. Bio-oil, a dark brown organic liquid, also called bio-fuel oil, pyrolytic oil, liquid wood, and pyrolysis oil, is produced by pyrolysis of biomass [7]. Pyrolysis involves the high thermal decomposition of organic material in the absence of oxygen (or any halogen). Chemical, biochemical, and thermal methods can be used to produce bio-oil from biomass. Woody biomass is often used to produce bio-oil. Traditional biomass (e.g., forest harvest residues, energy crops, and agricultural residues) comprises a high proportion (47%) of the renewable energy consumption [5]. Forest residue is one of the most geographically distributed biomass energy resources [8], and is attracting the attention of companies to produce bio-oil. One advantage of bio-oil is that it can be directly used in boilers as fuel [7]. In addition, utilizing

dead woody materials from forests to produce bio-oil offers a low-cost solution that can help bioenergy industries to grow, while also reducing forest fire risks [9]. According to the Oregon Department of forestry, more than 500,000 acres burned in Oregon in 2017 [10].

The biomass-to-bio-oil supply chain (BTBSC) consists of two sets of operations: conversion and logistics. The first set of operations involves all decisions relevant to activities for efficient conversion of biomass to bio-oil (e.g., process settings and production scheduling). The second set of operations, which we focus on here, focuses on improving decisions relevant to transportation and warehousing.

Recent growing interest within society over the use of renewable resources for bioenergy has stimulated research efforts in improving biomass processing solutions [11] and balancing the sustainability dimensions in logistics [12]. Mobile refineries have been proposed to overcome logistics challenges in BTBSCs, such as high handling and transportation costs due to biomass properties (bulky and low energy density). Mobile bio-refineries can enable more efficient transport systems. Instead of trucking biomass to a fixed bio-refinery, mobile bio-refinery units are transferred to the forest or farms to convert biomass feedstocks to intermediate products (e.g., bio-oil and bio-char). There are three main advantages of employing mobile refineries in BTBSCs [13]: (1) decreasing the total cost including fixed, variable, and labor costs; (2) reducing the required storage capacity for biomass; (3) reducing transportation costs. In particular, four factors have been identified that influence the benefits of employing mobile refineries: transportation distances, biomass type, policies and regulations, and time of the year.

Due to the challenges related to the use of fossil fuels noted above, it seems reasonable to understand the potential of alternative energy sources to reduce energy costs and carbon footprint. Prior research has investigated the effect of introducing mobile refineries into a conventional BTBSC by optimizing the supply chain based on logistics costs. However, research has not considered the impact of the cost of carbon on the structure and cost of mixed BTBSCs (employing fixed and mobile refineries). The objective of the research presented herein is to propose a mathematical modeling approach to simultaneously consider the factors of logistics and energy policy (carbon price) in the cost optimization of a mixed BTBSC. To complete such an optimization, the structure of the supply chain and its associated carbon footprint need to be understood.

In a BTBSC, there are three main sources contributing to carbon footprint: logistics, conversion processes, and employee commuting [14]. In addition to logistics costs, the cost of associated carbon emissions is considered here by extending prior research [13]. Specifically, this research builds on the prior work by integrating the cost of carbon into the mathematical cost model as a decision variable in the objective function. In addition, this work employs a metaheuristic method (a genetic algorithm) for the optimization of the proposed BTBSC, while the prior work used an exact optimization solver (Gurobi).

The research herein will enable the optimization of an improved mathematical model for BTBSCs utilizing mobile and fixed biorefineries, harvesting and collection sites, and distribution centers. The model includes the associated direct costs as well as the cost of carbon. This research contributes to the literature of bio-oil problems by

designing a genetic algorithm to find the best locations to establish mobile and fixed bio refineries for minimizing logistics and carbon costs.

The remainder of this paper is structured as follows: Sect. 2 provides a brief review of related work; Sect. 3 presents the assumptions required for modeling, as well as a description of the problem to be formulated. The solution algorithm is given in Sect. 4. Section 5 presents an application of the model. Finally, conclusions based on this work and opportunities for future research are given in Sect. 6.

2 Prior research

Two prior papers have reported comprehensive literature reviews on biomass-to-bioenergy supply chain optimization through 2016. Yue et al. [15] evaluated opportunities and key issues of BTBSCs with regard to sustainability factors, optimization methods, and mathematical models. Thereafter, Mirkouei et al. [16] carried out a fundamental review to examine approaches and methodologies used in BTBSC. For example, Zhu et al. [17] proposed a mixed integer linear programming (MILP) model for a numerical problem derived from the literature for the production of biofuel from biomass. Their problem involved two potential refinery locations, three potential storage locations, and ten biomass production fields. They solved the proposed mathematical model using CPLEX software. They considered one harvesting season (1 year of operation) in managing a four-echelon supply chain, which included biomass production, harvesting, biofuel production, and storage. With regard to the importance of using mobile refineries in BTBSCs, Mirkouei et al. [13] proposed a quantitative decision making approach. They presented an MILP model, solved by exact software (Gurobi), to obtain an optimal solution including the location of refineries, quantity of biomass and bio-oil, and a number of trips using a set of forest data from northwest Oregon. They showed that using mobile refineries leads to the reduction of system costs when transportations costs are high. In recent studies, while multi-objective problems have attracted much attention, use of mobile refineries has been ignored. Osmani and Zhang [12], for example, presented a multi-objective mathematical model with data derived from Wisconsin, in which the sale price, demand, and the amount raw materials were assumed to be probability parameters with known statistical distributions. Table 1 compares the research reported herein and the approached used in prior work. No prior studies were identified in the literature that designed a genetic algorithm for solving mobile refinery problems with carbon cost included in the total cost of BTBSCs. The next section presents a mathematical model to optimize logistics costs and environmental impacts in a renewable energy supply chain.

3 The proposed mathematical model

Most mathematical models used in transportation and logistics problems are formulated as mixed integer linear programming (MILP) problems. This section presents an MILP for a five-echelon BTBSC, including harvesting and collection sites, mobile and fixed refineries, and warehouses or distribution centers.

Table 1 Relevant studies

Studies	Model	CO_2	Solution algorithm	Mobile refinery	Carbon cost
Zhu et al. [17]	MILP	–	Optimizer solver: CPLEX	–	–
Mirkouei et al. [13]	MILP	–	Optimizer solver: Gurobi	Yes	–
Osmani and Zhang [12]	Stochastic MILP	Objective	Optimizer solver: GAMS	–	–
Present study	MILP	Variable	Metaheuristic approach: genetic algorithm	Yes	Yes

3.1 The problem statement

The five-echelon biomass-to-bioenergy network considered is shown in Fig. 1. To produce the final product (bio-oil) from woody biomass, the total cost includes logistics and carbon costs. Four truck types are assumed to be used in the network due to road restrictions: small tractor-trailers (st), large tractor-trailers (lt), small tanker trucks (stt), and large tanker trucks (ltt). Small tractor-trailers are used to transport biomass from harvesting sites to collection sites. They are also used to transport material from harvesting sites to mobile refineries. Biomass is sent from collection sites to mobile and fixed refineries using large tractor-trailers. Small tanker trucks are used to transport bio-oil from mobile refineries to fixed refineries, while bio-oil is transferred from fixed refineries to warehouses using large tanker trucks. Notably, this research considers the upstream and midstream segments in BTBSC, while the downstream segment, including distribution and demand activities, is left to future work. Since the problem was formulated in a deterministic environment, stochastic parameters such as demand should also be considered in the future models.

Several other factors must be considered in formulating the mathematical model. Biomass type, moisture content, and ash content, for example, affect production yield in biomass-to-bio-oil conversion. In addition, costs of refinery operation consist of transport and setup costs (mobile refineries only), feedstock handling cost, purchased energy cost, and repair and maintenance costs. The variable, fixed, and labor costs of harvesting, collection, and pre-processing activities consist of the associated costs of employing a harvester, forwarder, and grinder, respectively [13]. A third-party transportation business is assumed to be used. Further, the type of truck for each route and the available amount of biomass at each site are known. The time horizon is assumed to be 1 year.

3.2 The mathematical model

The mathematical model is presented below with regard to the BTBSC problem defined in Sect. 3.1. In this model, all parameters and variables are assumed to be

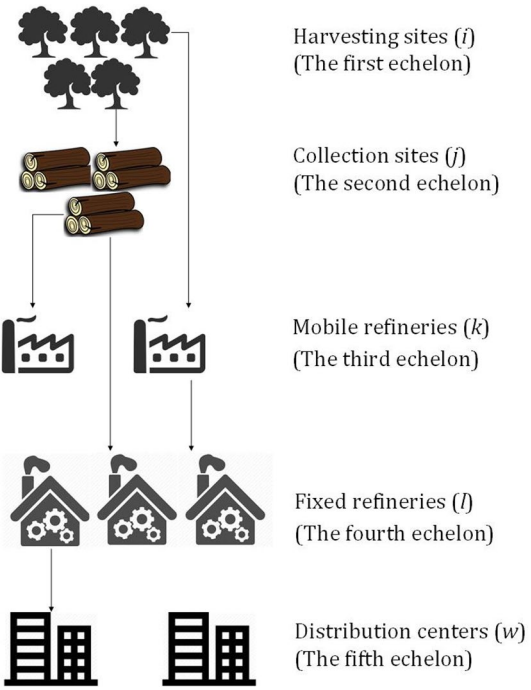


Fig. 1 Five-echelon biomass-to-bioenergy network

deterministic. The notations of indices, parameters, and variables used in the mathematical model are as follows.

Indices

i Index of harvesting sites, $i = 1,\ldots, I$
j Index of collection sites, $j = 1,\ldots, J$
k Index of mobile bio-refinery sites, $k = 1,\ldots, K$
l Index of fixed bio-refinery sites, $l = 1,\ldots, L$
t Index of time periods, $t = 1,\ldots, T$
w Index of warehousing sites, $w = 1,\ldots, W$

Parameters

a The lowest allowable level of biomass utilization
A_r Capacity of sites to store biomass or bio-oil, where $r \in \{j, k, l, \text{and } w\}$
c_v Vehicle capacity, where $v \in \{\text{st, stt, lt, and ltt}\}$
C_{rt} Total operation cost at time t for sites, where $r \in \{j, k, l, \text{and } w\}$
$D_{rr'}$ Distance between locations, where r and $r' \in \{j, k, l, \text{and } w\}$, while $r' \neq r$
e_v The full-load fuel consumption rate of vehicle, where $v \in \{\text{st, stt, lt, and ltt}\}$
$f_{rr't}$ The fuel consumption from site r to site r' at time t, where r and $r' \in \{j, k, l, \text{and } w\}$ while $r' \neq r$
M Large positive constant (also known as Big M)
P Percentage yield of converting biomass to bio-oil

p_v Cost per mile for vehicle, where $v \in \{\text{st, stt, lt, and ltt}\}$
u Carbon tax or price

Binary decision variables

L_{jt} Binary variable to select the location of collection sites at time t (1 indicates the location is selected and 0 indicates the location is not selected)
L_{kt} Binary variable to select the location of mobile refinery sites at time t
L_{lt} Binary variable to select the location of fixed refinery sites at time t
L_{wt} Binary variable to select the location of collection sites at time t

Decision variables

X_{ijt} Amount of biomass transported from site i to site j at time t
X_{ikt} Amount of biomass transported from site i to site k at time t
X_{jlt} Amount of biomass transported from site j to site l at time t
X_{jkt} Amount of biomass transported from site j to site k at time t
Y_{klt} Amount of bio-oil transported from site k to site l at time t
Y_{lwt} Amount of bio-oil transported from site l to site w at time t

Cost functions

FC Facility costs
TC Transportation costs
TCC Total carbon cost

The objective function includes two types of costs: logistics costs and carbon cost. Logistics costs include facility costs (Eq. 1) and transportation costs (shown in Eq. 2). The cost function is presented below in a piecewise manner.

The five primary activities considered (i.e., harvesting, collection, mobile refinery production, fixed refinery production, and warehousing) are completed in five different locations. The locations of harvesting sites and the mass of available raw material are known. However, the optimal locations for biomass collection, mobile refining, fixed refining, and warehousing are unknown. Thus, a set of four binary decision variables is used to determine the optimal location of each activity from the potential locations provided. To show the status of a location, a binary variable L is defined, where $L=1$ indicates the location is active, while $L=0$ indicates the location is inactive. With regard to total cost at time t for each location (C_{rt}), the facility costs (FC) are as follows (Eq. 1):

$$FC = \sum_{t=1}^{T} \sum_{j=1}^{J} \sum_{k=1}^{K} \sum_{l=1}^{L} \sum_{w=1}^{W} \left(C_{jt} L_{jt} + C_{kt} L_{kt} + C_{lt} L_{lt} + C_{wt} L_{wt} \right) \tag{1}$$

In order to calculate transportation cost, the routes and number of trips between locations must be known. The number of truck trips is calculated by rounding up

the total amount (mass or volume) of the product (biomass, X, and bio-oil, Y) being transported divided by the capacity of the truck (c) transporting the product between each location. With regard to distance between locations (D), cost per mile for trucks (p), and the amount of product being transported, the transportation cost (TC) is given as (Eq. 2):

$$TC = TC \text{ of biomass} + TC \text{ of bio-oil} \tag{2}$$

in which,

$$\text{TC of bio-oil} = \sum_{t}^{T}\left(\sum_{k}^{K}\sum_{l}^{L} p_{stt} D_{kl}\left\lceil \frac{Y_{klt}}{c_{stt}}\right\rceil + \sum_{l}^{L}\sum_{w}^{W} p_{ltt} D_{lw}\left\lceil \frac{Y_{lwt}}{c_{ltt}}\right\rceil\right) \tag{3}$$

$$\begin{aligned}\text{TC of biomass} = \sum_{t}^{T}\Bigg(&\sum_{i}^{I}\sum_{j}^{J} p_{st} D_{ij}\left\lceil \frac{X_{ijt}}{c_{st}}\right\rceil + \sum_{i}^{I}\sum_{k}^{K} p_{st} D_{ik}\left\lceil \frac{X_{ikt}}{c_{st}}\right\rceil \\ &+ \sum_{j}^{J}\sum_{k}^{K} p_{lt} D_{jk}\left\lceil \frac{X_{jkt}}{c_{lt}}\right\rceil + \sum_{j}^{J}\sum_{l}^{L} p_{lt} D_{jl}\left\lceil \frac{X_{jlt}}{c_{lt}}\right\rceil\Bigg)\end{aligned} \tag{4}$$

In this model, carbon cost is calculated assuming a carbon tax rate (\$/kg CO_2) multiplied by the mass of CO_2 emissions generated by the trucks. It can be noted, "A carbon tax is a fee for making users of fossil fuels pay for climate damage their fuel use imposes by releasing carbon dioxide into the atmosphere, and for motivating switches to clean energy" [18]. Total CO_2 emissions caused by the transportation network are calculated using a constant-speed approach [19, 20]. Generally, three parameters are used to define a truck's fuel consumption: speed, mass of load, and travel distance [19]. If the truck speed is constant, the fuel consumption from the ith node to the jth node in the tth period, f_{ijt}, without considering the return route can be estimated as follows [19, 20]:

$$f_{ijt} = D_{ij} e_{st} X_{ijt} / c_{st} \tag{5}$$

in which, e_{st} indicates the full-load fuel consumption rate of a small tractor-trailer.

If u indicates the carbon tax, the total carbon cost (TCC) is predicted as:

$$TCC = u\sum_{t}^{T}\left(\sum_{i}^{I}\sum_{j}^{J} f_{ijt} + \sum_{i}^{I}\sum_{k}^{K} f_{ikt} + \sum_{j}^{J}\sum_{k}^{K} f_{jkt} + \sum_{j}^{J}\sum_{l}^{L} f_{jlt} + \sum_{k}^{K}\sum_{l}^{L} f_{klt} + \sum_{l}^{L}\sum_{w}^{W} f_{lwt}\right) \tag{6}$$

According to the aforementioned equations related to costs, the objective function to minimize total cost is (Eq. 7):

$$Min\, TC = FC + TC + TCC \tag{7}$$

Subject to (Eqs. 8–21):

$$\sum_{l\in L} X_{jlt} + \sum_{k\in K} X_{jkt} = \sum_{i\in I} X_{ijt} \quad \forall j \in J, t \in T \tag{8}$$

$$\sum_{i\in I} Y_{klt} = P \times \sum_{i\in I} X_{ikt} \quad \forall k \in K, t \in T \tag{9}$$

$$\sum_{w\in W} Y_{lwt} = P \times \sum_{j\in J} X_{jlt} \quad \forall l \in L, t \in T \tag{10}$$

$$\sum_{i\in I} X_{ijt} \le M \times L_{jt} \quad \forall j \in J, t \in T \tag{11}$$

$$\sum_{i\in I} X_{ikt} + \sum_{j\in J} X_{jkt} \le M \times L_{kt} \quad \forall k \in K, t \in T \tag{12}$$

$$\sum_{j\in J} X_{jlt} + \sum_{k\in K} Y_{klt} \le M \times L_{lt} \quad \forall l \in L, t \in T \tag{13}$$

$$\sum_{l\in L} Y_{lwt} \le M \times L_{wt} \quad \forall w \in W, t \in T \tag{14}$$

$$\sum_{i\in I} X_{ijt} \le A_j \quad \forall j \in J, t \in T \tag{15}$$

$$\sum_{i\in I} X_{ikt} + \sum_{j\in J} X_{jkt} \le A_k \quad \forall k \in K, t \in T \tag{16}$$

$$\sum_{j\in J} X_{jlt} \le A_l \quad \forall l \in L, t \in T \tag{17}$$

$$\sum_{l\in L} \left(Y_{lwt} + \sum_{k\in K} Y_{klt} \right) \le A_w \quad \forall w \in W, t \in T \tag{18}$$

$$\sum_{t\in T} \left(\sum_{i\in I} \sum_{j\in J} X_{ijt} + \sum_{i\in I} \sum_{k\in K} X_{ikt} \right) \ge a \tag{19}$$

$$X_{ijt}, X_{ikt}, X_{jkt}, X_{jlt}, Y_{klt}, Y_{lwt} \ge 0 \quad \forall i \in I, j \in J, l \in L, k \in K, w \in W, t \in T \tag{20}$$

$$L_{jt},\ L_{kt},\ L_{lt}, L_{wt} = \begin{cases} 1 & \text{if location is open,} \\ 0 & \text{otherwise.} \end{cases} \quad \forall j \in J, k \in K, l \in L, w \in W, t \in T \tag{21}$$

Equation 8 represents the conservation of flow in and out of node j. Equations 9–10 present the conversion rate of biomass to bio-oil. Equations 11–14 show the status of locations (active or inactive) (M is a large positive constant). Equations 15–18 show

the capacity constraint for each location. Equation 19 indicates that the available amount of biomass in all harvesting sites should be used. Equations 20–21 present the decision variables. Section 4 proposes a genetic algorithm to find the decision variables X_{ijt}, X_{ikt}, X_{jkt}, X_{jlt}, Y_{klt}, Y_{lwt}, L_{jt}, L_{kt}, L_{lt}, and L_{wt} for optimizing the objective function presented in Eq. 7 with regard to the model constraints.

4 Solution algorithm: meta-heuristic optimization

MILP models for large scale problems are unable to be solved by exact methods or solver software due to their complexity, which is NP hard [21]. Metaheuristic approaches, which are also called approximate algorithms [22], are the most powerful for optimizing MILP models [23]. Metaheuristic algorithms are classified as single-based search, which start with a single candidate solution, and population-based search, which start with a population of candidate solutions [23]. Population-based methods usually outperform single-based search methods in speed and accuracy [22]. To solve the mathematical model presented here, therefore, a genetic algorithm (GA) is employed. GA is the most powerful population-based method for solving MILP models [24].

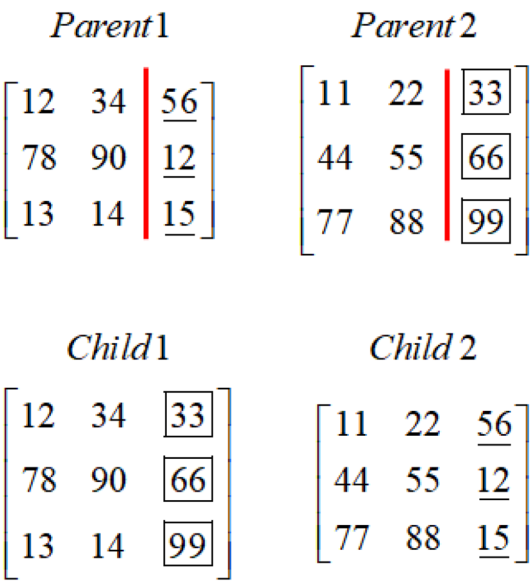
In GA terminology, a chromosome, which is a set of genes, a candidate for the solution is called. A chromosome consists of the decision variables. The genetic algorithm creates a random chromosome in a population as the solution of the problem, and then improves the solution with operators named "crossover" and "mutation" in a defined iteration [23]. In this research, the GA parameters are the probability of "two-point crossover" and "random mutation", population size, and the number of iteration steps. The chromosome of this problem is designed as a matrix, including the decision variables. The GA is developed by language programming of MATLAB 2017b, and solved using a Windows 10 64-bit Operation System with Intel Core i5 processor (CPU 3.40 GHz) and 16 GB RAM.

The chromosome matrix includes the following decision variables: X_{ijt}, X_{ikt}, X_{jkt}, X_{jlt}, Y_{klt}, Y_{lwt}, L_{jt}, L_{kt}, L_{lt}, and L_{wt} which utilize direct values, rather than being coded. Equation 22 shows the X_{ijt} chromosome for four harvesting sites (rows) and five collection sites (columns).

$$X_{ijt} = \begin{bmatrix} X_{11} & X_{12} & X_{13} & X_{14} & X_{15} \\ X_{21} & X_{22} & X_{23} & X_{24} & X_{25} \\ X_{31} & X_{32} & X_{33} & X_{34} & X_{35} \\ X_{41} & X_{42} & X_{43} & X_{44} & X_{45} \end{bmatrix}_t = \begin{bmatrix} 13 & 63 & 12 & 0 & 9 \\ 25 & 43 & 23 & 63 & 22 \\ 13 & 122 & 212 & 434 & 545 \\ 123 & 20 & 40 & 100 & 500 \end{bmatrix}_t \tag{22}$$

As noted above, the process of creating two parent solutions and then producing a child solution using both is called *crossover*. There are several methods to perform crossover: single-point crossover, two-point crossover, uniform crossover, half uniform crossover, and three-parent crossover, to name a few. In this research, one-point crossover is used to create new generations, as shown in Fig. 2. The vertical lines in the parent matrices represent the points of crossover. The selected genes,

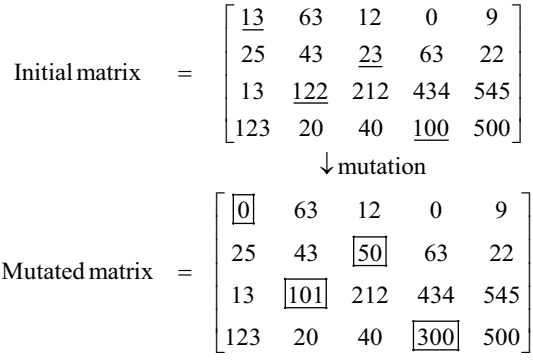
Fig. 2 Schematic of the single-point crossover method



which are to the right of the vertical lines and indicated using underlined and boxed text, are swapped between the parents, creating the two children.

As noted above, another operator used in GA to create a new generation is *mutation*. The mutation operator is employed to protect the diversity from one generation to the next. One or more gene values in a chromosome are selected and modified in the mutation process to generate a new solution. By using mutation, the performance of GA can be improved by avoiding locally-optimal solutions. There are different mutation types, including "bit string", "flip bit", "boundary", "non-uniform", "uniform", and "Gaussian" [25]. This research employs the uniform mutation method to create new generations, as shown in Fig. 3. Genes are randomly selected (underlined genes in initial matrix), and then recreated to generate a new chromosome (boxed genes in the mutated matrix).

Fig. 3 Schematic of the uniform mutation method



To select chromosomes from among the population as parents for crossover or as candidates for mutation, there are several selection methods, including "simplex crossover", "tournament", and "rank" selection [25]. This research uses tournament selection to choose an individual chromosome from the population, due to suitable coding efficiency [26]. Pseudo code for tournament selection method is as follows:

- Randomly select g individuals (the tournament size) from the population,
- Select the best individual from the pool, with probability q (between 0 and 1),
- Select the second best individual, with probability $q \times (1-q)$,
- Select the third best individual, with probability $q \times (1-q)^2$.

A constrained opimization problem, such as used herein, can be solved with GA using penalty methods. Penalty methods convert constrained problems into unconstrained problems, and the provided solutions are valid for the original constrained problem. This research uses the death penalty method to obtain feasible solutions for the constrained problem. The death penalty method does not discard solutions, rather it introduces a penalty into the objective function when the solution obtained does not satisfy the constraints. The obtained solution is penalized by multiplying the constraint violations by penalty parameters.

5 Application of the model

In this section, small and large test problems are first presented to demonstrate the convergence of the model, and then a sensitivity analysis of bio-refinery capacity is conducted to test the behavior model.

5.1 Small and large test cases

In this section, two test problems (small and large) are solved with regard to the designed chromosome to be used in the GA. The test problems are used to evaluate the ability of the GA to repeatably converge to a near-optimal solution. The small test problem includes three harvesting sites, five collection sites, seven potential mobile refinery sites, eight potential fixed refinery sites, and two distribution sites. The large test problem includes twenty harvesting sites, eight collection sites, six potential mobile refinery sites, four potential fixed refinery sites, and two distribution sites. The latitude and longitude values for the potential locations were randomly generated. It is assumed that the four types of trucks noted above are available for transporting biomass and bio-oil. The mathematical model employs six sets of continuous decision variables, four sets of binary decision variables, and 14 sets of constraints for a set of locations, as defined by the general formulation provided above.

The technical parameter values used in metaheuristic algorithms affect the solver performance and the solution obtained. For the test problems, the values of the four parameters used, population, probability of crossover, probability of mutation, and

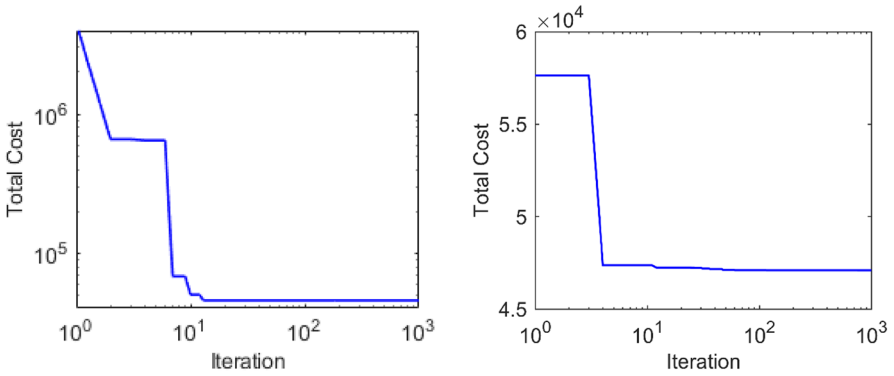


Fig. 4 Lowest total cost obtained for each iteration (indicating convergence of the GA): small test problem (left) and large test problem (right)

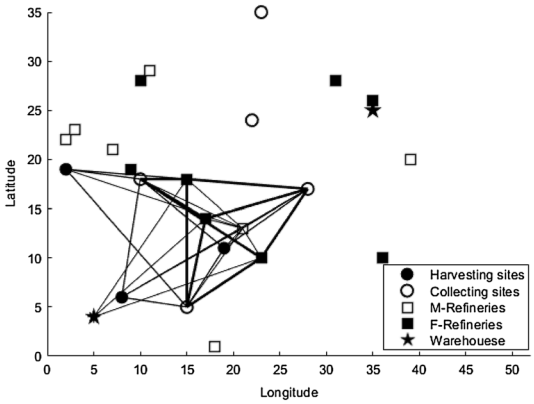


Fig. 5 The identified optimal network (other potential locations for the small test problem also indicated)

number of iterations are 100, 0.8, 0.05, and 1000 respectively. These values were tuned by trial and error, which was more appropriate for the test problems than using published values. For the small and large test problems, the lowest total cost ("best cost") for each iteration is presented in Fig. 4.

It can be concluded that the GA converges to provide a near-optimal solution for the model presented in Eq. 7. There is not a significant reduction in the total cost obtained after the 140th iteration. Figure 5 shows the lowest cost network found by the GA for the small bio-oil supply chain, which includes three collection sites, one mobile-refinery, three fixed-refineries, and one warehouse. These optimal locations were selected from the set of potential locations (three harvesting sites, five collection sites, seven potential mobile refinery sites, eight potential fixed refinery sites, and two distribution sites), which are also indicated in the figure.

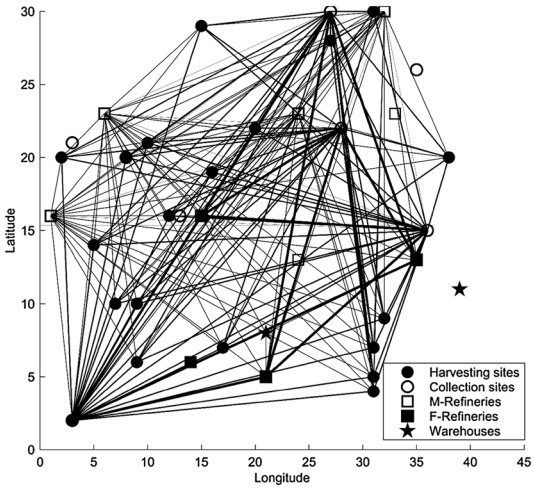


Fig. 6 The optimal solution presented by GA with the potential locations for the large test problem

Figure 6 shows the lowest cost network found by the GA for the large bio-oil supply chain, which is comprised of four collection sites, four mobile-refinery, three fixed-refineries, and one warehouse. These locations were optimally selected from the set of potential locations (twenty harvesting sites, eight collection sites, six potential mobile refinery sites, four potential fixed refinery sites, and two distribution sites), which are also indicated in the figure.

5.2 Sensitivity analysis

Sensitivity analysis is performed to study the effect of the selected mathematical model inputs, which can vary, on the output of the model. Different scenarios are considered by varying model parameter values to observe the behavior of model. In general, when no constraints are placed on the capacities of harvesting and collection sites, mobile and fixed refineries, and warehouses, only one location will be selected from each set of collection, fixed refinery, mobile refinery, and warehouse locations. This result shows the expected behavior of the proposed mathematical model to optimize total cost by reducing the cost of active centers (selecting one location for processing and one location for storage). When we increase facility cost, fewer facilities will be selected. The total cost increases as a result of increasing cost parameters, e.g., transportation costs, carbon tax, and operating costs. Tables 2, 3 report the different scenarios for the capacity of refineries. Scenario 1 in Table 2 indicates a case in which a refinery has a capacity such that is able to store all of the incoming biomass from the harvesting and collection sites. In Scenario 2, the total cost and the number of active mobile refineries increase when mobile refinery capacity decreases by 35%. The same characteristic is seen for Scenario 3 and 4.

With regard to the original case (Scenario 1) in Table 3, Scenario 5 indicates the total cost and the number of fixed refineries increase when fixed refinery capacity decreases by 76%. When one fixed refinery does not have sufficient capacity to process all biomass, the bio-oil supply chain will require additional fixed refineries

Table 2 Four scenarios exploring the effect of mobile refinery capacity on supply chain cost and active entities

Scenario	Capacity	Variability in capacity (%)	Variability in costs (%)	AMR	AFR	ACS	AWS
1	850	0	0	1	1	1	1
2	550	−35	14	2	1	1	1
3	300	−65	36	3	1	2	1
4	200	−76	77	5	1	3	1

AMR active mobile-refinery, *AFR* active fixed-refinery, *ACS* active collection site, *AWS* active warehouse site

Table 3 Three scenarios for the capacity of fixed refineries

Scenario	Capacity	Variability in capacity (%)	Variability in costs (%)	AMR	AFR	ACS	AWS
0	850	0	0	1	1	1	1
1	200	−76	35	1	2	2	1
2	130	−85	57	2	2	3	1

AMR active mobile-refinery, *AFR* active fixed-refinery, *ACS* active collection site, *AWS* active warehouse site

until the capacity is met, resulting in a cost increase (35% increase in Scenario 1). The same characteristic is seen for Scenario 2 (57% cost increase for a decrease in capacity of 85%).

According to the results of the sensitivity analysis, it can be said that the mathematical model presented in Eq. 7 has reasonable relationships between input and output variables. No errors with the model were identified, as no unexpected behaviors were revealed during the sensitivity analysis.

5.3 Managerial implications

The findings from this study should aid managers in planning for supply chains in which biomass is used to produce bio-oil. In particular, logistics decision makers should take note of the importance of mobile refineries in the production of bio-oil, which can be a starting point for establishing a cost-effective, environmentally responsible bio-oil supply chain. Our research finds that mobile refineries are able to appreciably reduce the total cost and negative environmental impacts of bio-oil production, compared to fixed refineries. In bio-oil supply chains, managers should play an active role in ensuring that bio-oil is cost-competitive in the market, while also generating lower environmental impacts than fossil-based products. Thus, the proposed mathematical model acts as a decision tool for managers, even for large-scale problems. The available solvers proposed in the literature review would fail to provide a solution for bio-oil supply chain problems involving a high number (hundreds

to thousands) of forest harvesting sites. Our finding shows that the proposed algorithm is able to provide a solution for large size problems.

6 Conclusion

More than 500,000 acres of forests burned in Oregon in 2017, presumed to be due to changes in climate and forest management practices. Increased collection of dying and dead woody biomass from forests could help reduce fire hazards. This research drew inspiration from this impactful problem. A mathematical modeling approach was presented for optimizing the total cost of a biomass to bio-oil supply chain. The mathematical model was based on a five echelon supply chain comprised of harvesting sites, collection sites, mobile refineries, fixed refineries, and warehouses. To extend prior related research, the model presented here relaxed a limitation of mathematical models for bio-oil supply chain problems by considering carbon cost as a part of the total supply chain cost. The aim of mathematical cost modeling was to find a near-optimal solution, including the number of truck trips, mass of biomass processed, mass of bio-oil produced, number of mobile and fixed refineries, and number of warehouses. Since the proposed model is a mixed integer linear programming problem, a meta-heuristic optimization approach (genetic algorithm) was designed to find an optimized solution.

By evaluating two test cases (small and large) and conducting a sensitivity analysis, it was shown that the proposed mathematical model can be applied in the design of a biomass to bio-oil supply chain including mobile refineries. Decision makers will be able to select the optimal number of mobile and fixed refineries with regard to total cost, which consists of logistics cost and carbon cost. Based on the presented approach, the mass of carbon emissions is first estimated to calculate the related carbon tax. Since the proposed methodology uses a genetic algorithm, it can be applied for large scale problems to overcome restrictions of exact methods.

Several limitations of mathematical modeling for biomass to bio-oil supply chains should be addressed by future research. The model presented herein was formulated in a deterministic environment. As shown in the sensitivity analysis, the proposed model does not have robustness in the presence of uncertainty. To address this limitation, stochastic parameters could be included in the mathematical model. Alternatively, design of experiments approaches, such as the Taguchi method, would aid in better calibrating the genetic algorithm parameters than tuning through trial and error. As a key limitation of this research to perform more comprehensive sustainability assessment, only economic (supply chain cost) and environmental (carbon emissions) were included in the model, both represented as costs. The solution obtained by the genetic algorithm was optimized without considering broader economic, environmental, and social effects. Future research should address extensions of the mathematical model to simultaneously improve economic, environmental, and social performance of bioenergy supply chains.

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